

AURORA, NE, USA

WMO: 725513

Lat: 40.893N

Lon: 97.997W

Elev: 1799

StdP: 13.77

Time Zone: -6.00 (NAC)

Period: 98-19

WBAN: 04901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.7	2.9	-9.2	3.6	-0.1	-4.7	4.6	4.9	35.1	29.4	32.4	25.4	9.7	330	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	93.2	73.8	90.6	73.0	88.1	72.3	78.1	87.5	76.4	85.9	74.8	84.1	12.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.3	141.8	83.4	73.3	132.3	81.6	72.1	127.2	80.4	43.1	87.5	41.1	86.3	39.5	83.6	84.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
30.1	26.2	23.3	-10.5	98.8	3.5	2.5	-13.0	100.7	-15.1	102.1	-17.1	103.6	-19.6	105.4		
			DB													
			WB	-11.1	81.8	3.4	1.6	-13.5	83.0	-15.5	83.9	-17.4	84.9	-19.9	86.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	26.1	28.3	40.5	50.8	61.6	71.3	75.4	65.7	52.5	39.9	28.2
		DBStd	19.93	11.67	12.23	11.16	9.70	8.79	6.88	5.60	5.51	8.12	9.33	10.37
		HDD50	2886	741	609	334	105	12	0	0	2	81	327	675
		HDD65	6009	1206	1027	762	434	171	22	3	88	398	752	1140
		CDD50	3332	0	2	38	129	372	638	787	475	159	25	0
		CDD65	982	0	0	1	9	66	210	325	249	111	0	0
		CDH74	9845	0	0	33	164	774	2148	3200	2186	1168	163	9
		CDH80	3755	0	0	6	43	260	832	1332	818	421	42	1
(14)	Wind	WSAvg	10.7	11.2	11.6	12.3	13.6	12.2	10.4	7.8	7.4	9.0	10.3	11.1
(15)	Precipitation	PrecAvg	27.7	0.6	0.7	1.4	2.5	4.6	4.4	3.6	3.4	2.3	2.1	1.3
(16)		PrecMax	38.1	1.5	1.7	3.2	4.4	8.5	7.7	8.3	8.1	4.8	6.8	3.9
(17)		PrecMin	16.3	0.1	0.0	0.0	0.3	1.4	1.5	1.2	1.1	0.4	0.0	0.0
(18)		PrecStd	5.0	0.4	0.4	0.9	1.0	1.9	1.8	1.9	1.7	1.3	1.8	0.9
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	61.1	66.5	78.4	85.7	91.3	95.7	97.2	96.6	93.1	86.0	74.0	60.0
(20)		0.4% MCWB	45.0	49.6	57.3	62.3	65.4	71.7	76.0	73.4	70.1	64.8	55.0	45.3
(21)		2% DB	51.8	57.4	71.5	78.9	86.4	91.3	93.2	91.0	89.4	78.6	67.7	53.5
(22)		2% MCWB	41.3	45.7	54.6	59.1	65.6	71.8	76.3	74.0	69.7	59.7	52.1	44.6
(23)		5% DB	45.9	51.7	64.9	73.0	82.1	88.4	90.5	88.1	85.2	73.2	61.5	47.9
(24)		5% MCWB	38.3	43.6	52.3	56.7	64.1	71.2	75.5	74.2	68.4	57.7	49.9	40.5
(25)		10% DB	41.1	45.1	59.2	68.2	78.2	84.9	87.9	84.3	81.2	69.0	55.9	43.1
(26)		10% MCWB	35.7	38.9	48.9	54.8	62.3	69.6	74.5	72.7	65.9	56.3	46.4	37.0
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	45.9	52.7	62.0	64.8	71.7	77.5	81.2	79.7	74.8	68.3	58.8	51.2
(28)		0.4% MCDB	57.8	59.4	69.6	77.8	82.6	87.5	90.1	87.4	84.1	78.3	66.5	57.6
(29)		2% WB	41.9	47.1	58.7	61.6	69.1	75.2	79.0	77.5	72.7	64.0	54.6	44.3
(30)		2% MCDB	50.5	57.3	66.2	71.6	80.5	86.0	88.6	86.7	83.0	71.6	64.0	52.3
(31)		5% WB	38.9	42.8	53.6	59.1	67.1	73.4	77.2	75.6	70.9	60.9	50.9	40.8
(32)		5% MCDB	45.2	51.1	62.8	70.5	77.8	84.2	86.8	84.8	80.8	68.6	60.0	47.3
(33)		10% WB	36.0	39.0	49.2	56.1	64.7	71.5	75.5	73.9	68.8	57.7	47.2	37.3
(34)		10% MCDB	40.8	45.1	59.4	67.0	74.5	82.1	84.8	82.2	77.7	66.7	56.6	42.6
(35)	Mean Daily Temperature Range	MDBR	19.1	19.6	22.2	23.2	21.9	21.2	20.1	20.3	23.8	23.3	22.3	18.8
(36)		5% DB MCDBR	26.6	29.4	32.5	31.9	28.1	25.2	23.6	24.3	27.2	30.3	30.8	26.6
(37)		5% WB MCWBR	17.9	18.5	18.1	16.5	11.8	10.7	11.5	11.1	11.5	15.0	17.8	18.2
(38)		5% DB MCDBR	24.9	28.1	28.6	27.2	24.0	22.3	20.9	21.2	23.2	22.8	27.8	24.5
(39)		5% WB MCWBR	17.1	18.3	17.3	16.0	12.0	11.2	11.7	11.6	11.5	13.8	17.2	17.6
(40)	Clear-Sky Solar Irradiance	taub	0.275	0.282	0.310	0.347	0.372	0.385	0.401	0.399	0.364	0.323	0.283	0.273
(41)		taud	2.481	2.493	2.445	2.360	2.363	2.378	2.342	2.342	2.396	2.495	2.521	2.492
(42)		Ebn at Noon	284	299	300	293	287	282	276	274	277	279	279	276
(43)		Edh at Noon	24	27	32	37	38	38	39	38	34	27	23	22
(44)	All-Sky Solar Radiation	RadAvg	693	964	1289	1630	1896	2161	2128	1849	1503	1076	767	584
(45)		RadStd	62	65	108	93	120	134	108	108	95	109	54	58

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page